

Towards Improved Uncertainty Quantification of Stochastic Epidemic Models Using Sequential Monte Carlo

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JSM 2024

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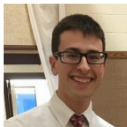
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■ Context

- Broader: Sequential calibration of stochastic computer models
- Focused: Tracking time-varying epi-dynamics, imperfect data, multiple sources of uncertainty, fast time to solution

■ Approach

- Trajectory-oriented calibration framework
- Sequential Monte-carlo for parameter inference

■ Illustration

- Covid-19 application

Broader Context

■ Computer simulation

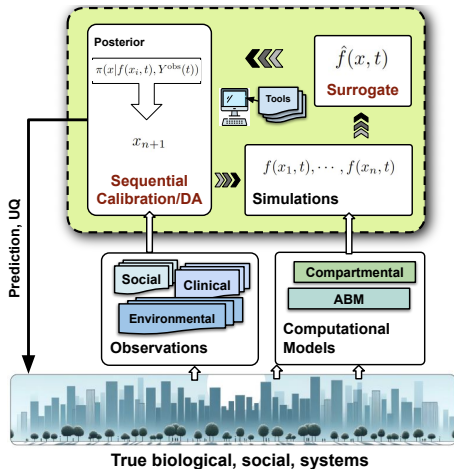
Encode the underlying epidemic process to study and predict its behavior.

■ Stochastic model

Encode uncertainties in underlying processes through randomness (e.g., **seeding infections, specific mixing patterns**).

■ Statistical Inference

Tools to calibrate the model to empirical reality, extra attention needed for stochastic model.



Focused Context

Computer simulation

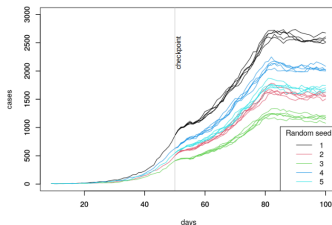
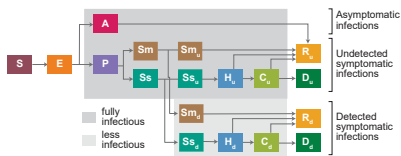
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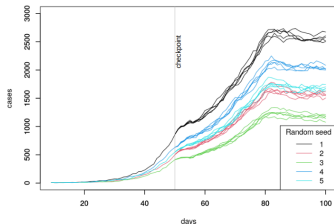
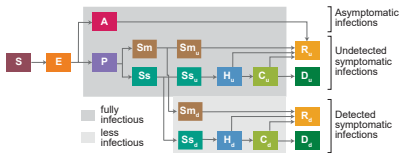
Some aspects of the simulation model

■ Each trajectory determines an unique pattern

Combination of parameter input and **random seed** (that controls the randomness) defines a specific history with probabilistic infections, disease progression, and population mixing in the heterogeneous contact network.

■ Checkpointing

It is used to serialize simulation states at specified times and then restart them from those times to continue forward.



Calibration objectives and challenges

The model parameters need to be estimated **periodically** from the empirical data.

■ Challenges

- Stochastic simulations.
- Parameters can vary over time (e.g., disease transmissibility).
- Some reported data are biased (e.g., under-ascertainment (reporting), delays in reporting).
- Model needs to be readjusted often to reflect the changing dynamics (e.g., new variant)
- The simulation history needs to be preserved to track unique population mixing into the future.

Trajectory-oriented Sequential Monte-Carlo

- 1 **Model linking simulation to reported data:** A **bias** model to connect the realizations of the stochastic simulation to the data which may contain biases (e.g., under-reporting, missing response).
- 2 **Sequential (in time) exploration of ~~optimal~~ parameters trajectories:** several data-assimilation steps that are executed sequentially in time.
 - In each data-assimilation window, importance sampling is used to estimate the model parameters.
 - The updated posterior samples are used as prior for the next time-window.
 - Use **random seed** as a paramter.

Connecting models to data

■ Calibration model

$$\mathbf{y}_{1:T} = \boldsymbol{\eta}_{1:T}(\boldsymbol{\theta}, s) + \boldsymbol{\varepsilon}_{1:T}, \quad \boldsymbol{\varepsilon}_{1:T} \sim \text{MVN}(\mathbf{0}, \boldsymbol{\Sigma})$$

■ Incorporating data bias

$$\mathbf{y}_t = \eta_t^{\text{obs}}(\boldsymbol{\theta}, s, \rho_t) + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_t^2), 0 < \rho_t < 1$$

$$\eta_t^{\text{obs}}(\boldsymbol{\theta}, s, \rho_t) \sim \text{Binomial}(\eta_t(\boldsymbol{\theta}, s), \rho_t)$$

Connecting models to data

■ Calibration model

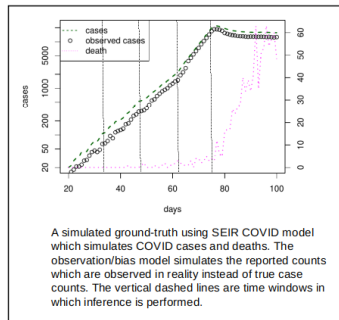
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Require: N (total simulation budget)

- 1: **Sample:** $(\theta_i, s_i, \rho_i) \sim \pi(\theta, s, \rho), \quad i = 1 : N.$
- 2: **Run simulation:** $\eta_{1:t_1}^c(\theta_i, s_i, \rho_i), \eta_{1:t_1}^d(\theta_i, s_i, \rho_i).$
- 3: **Compute weight:**

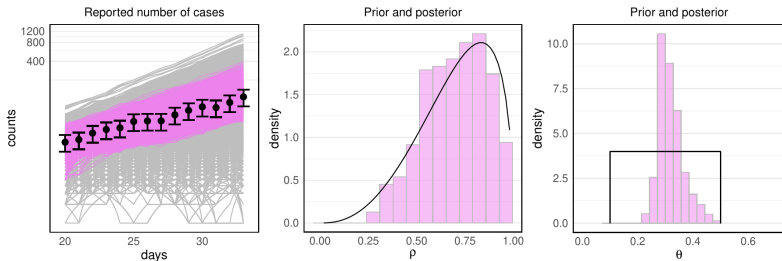
$$w_i^{(t_1)} \propto l(\mathbf{y}_{1:t_1}^c | \theta_i, s_i, \rho_i) \times l(\mathbf{y}_{1:t_1}^d | \theta_i, s_i, \rho_i)$$

- 4: **Resample:** Sample $(\theta_1, s_1, \rho_1), \dots, (\theta_N, s_N, \rho_N)$ with probabilities proportional to $w_i^{(t_1)}$.

Sequential posterior

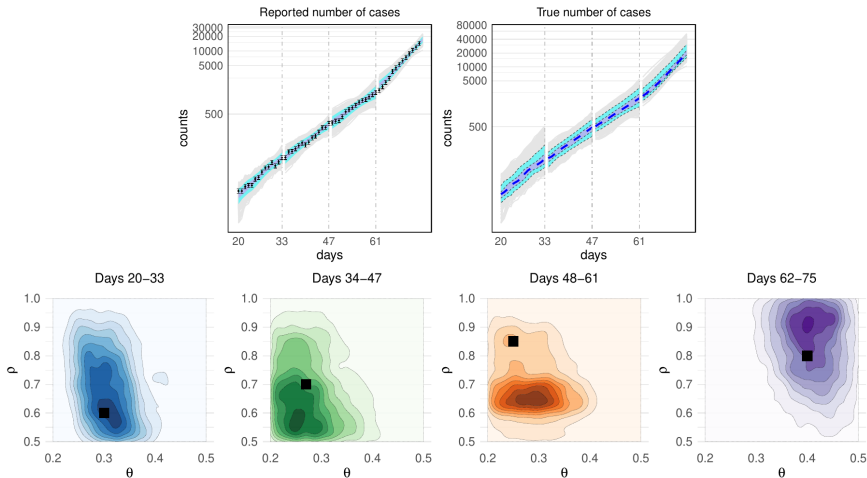
$$\begin{aligned}\pi(\theta, s, \rho | \mathbf{y}_{1:t_2}^c, \mathbf{y}_{1:t_2}^d) &= \frac{l(\mathbf{y}_{1:t_2}^c, \mathbf{y}_{1:t_2}^d | \theta, s, \rho) \pi(\theta, s, \rho)}{\pi(\mathbf{y}_{1:t_2}^c, \mathbf{y}_{1:t_2}^d)} \\&= \frac{l(\mathbf{y}_{1:t_1}^c, \mathbf{y}_{1:t_1}^d | \theta, s, \rho) \pi(\theta, s, \rho)}{\pi(\mathbf{y}_{1:t_1}^c, \mathbf{y}_{1:t_1}^d)} \times \\&\quad \frac{l(\mathbf{y}_{(t_1+1):t_2}^c, \mathbf{y}_{(t_1+1):t_2}^d | \mathbf{y}_{1:t_1}^c, \mathbf{y}_{1:t_1}^d, \theta, s, \rho)}{\pi(\mathbf{y}_{(t_1+1):t_2}^c, \mathbf{y}_{(t_1+1):t_2}^d | \mathbf{y}_{1:t_1}^c, \mathbf{y}_{1:t_1}^d)} \\&\propto \pi(\theta, s, \rho | \mathbf{y}_{1:t_1}^c, \mathbf{y}_{1:t_1}^d) \times \\&\quad l(\mathbf{y}_{(t_1+1):t_2}^c, \mathbf{y}_{(t_1+1):t_2}^d | \mathbf{y}_{1:t_1}^c, \mathbf{y}_{1:t_1}^d, \theta, s, \rho)\end{aligned}$$

Some results



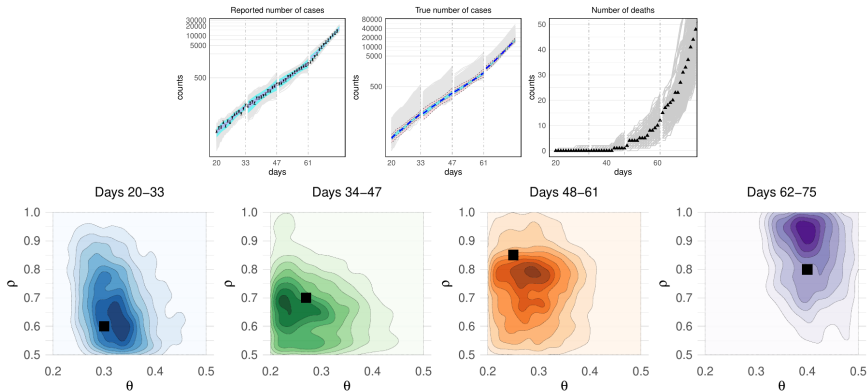
Calibration using reported case counts within the first time window –
(left) The simulated ground truth is shown by black dots which are used to find the posterior trajectories (in purple) that are consistent with the ground truth from the prior trajectories (in grey), (center) the prior density (in black) and posterior of the ρ parameter in the binomial bias model, (right) the prior and posterior distribution of simulation input θ .

Sequential calibration



Sequential calibration using reported case counts across four time windows

Sequential calibration



Sequential calibration using reported case counts and deaths across four time windows

- Easy access to uncertainties and ability to incorporate multiple data streams (multi-objective).
- Ability to account for data biases, varying parameter values throughout the pandemic.
- Uses simulation model directly inside the SMC framework, lacks scalability.

Surrogate-based approach: As an alternative, we have developed a Gaussian process based surrogate (CRNGP) that uses random seed as an input and emulates each trajectory.

References

- Fadikar, Arindam, Nicholson Collier, Abby Stevens, Jonathan Ozik, Mickaël Binois, and Kok Ben Toh. “Trajectory-oriented optimization of stochastic epidemiological models.” In 2023 Winter Simulation Conference (WSC), pp. 1244-1255. IEEE, 2023.
- Fadikar, Arindam et al., “Towards Improved Uncertainty Quantification of Stochastic Epidemic Models Using Sequential Monte Carlo,” in 2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), San Francisco, CA, USA, 2024 pp. 843-852. doi: 10.1109/IPDPSW63119.2024.00151

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